

PRD • HACKATHON ENTRY 2026

SmartTutor AI

Personalized Learning & Doubt-Solving Agent Orchestrated with Mastra, Qdrant & Enkrypt AI

TEAM

AI Innovators

EVENT

HiDevs × Mastra Hackathon

TECH STACK

Mastra + Qdrant + Enkrypt AI

The Learning Dilemma & Solution

! Core Problem

Students wrestle with generic, flat learning environments that fail to cater to individual speeds or remember session context. Standard chatbots act as temporary, state-less answer boxes rather than long-term mentors.

Key friction: Lack of memory, static test formats, and no adaptive safety rails.

✓ Proposed Solution

SmartTutor AI establishes a persistent cognitive tutor. Orchestrated via smart workflows, it learns the student's exact style, proactively tests weak concepts, schedules tasks, and ensures rigorous safety verification.

Ultimate outcome: High-grade, memory-retaining, and educational-safe academic partner.

Target Audience & Student Persona



Meet Ritesh: Primary User Persona

Profile: First-Year Data Science Student navigating complex Python libraries and core Data Structures & Algorithms (DSA).

Primary Goals: Realize an intuitive pathway to write cleaner code, retain hard memory structures, and consistently review learning blocks without stress.

Friction Points: Struggles with standard static videos; requires adaptive feedback, concept reinforcement, and dynamic revision intervals tailored to performance.

The Core AI Agent Ecosystem



Adaptive Tutor

Explains complex computer science & school concepts uniquely matched to the student's background level.



Dynamic Quiz

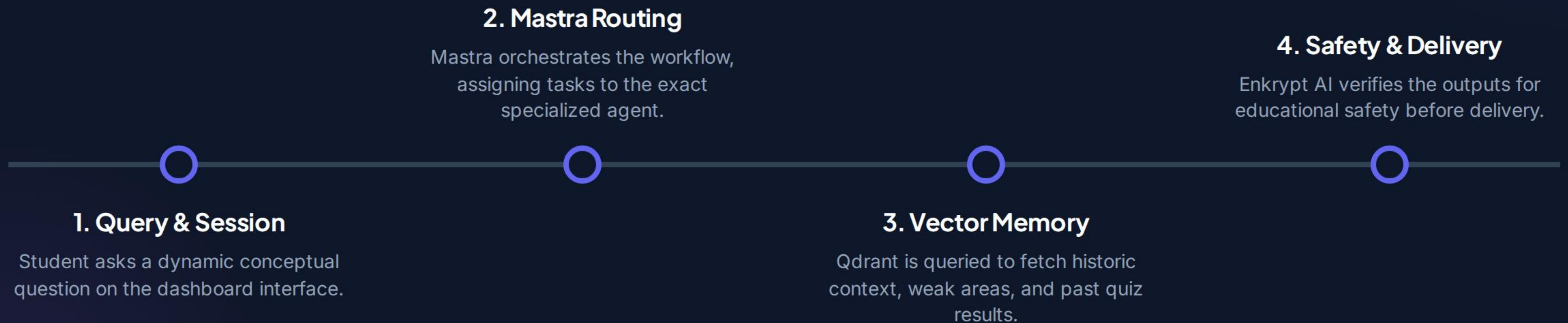
Generates targeted MCQs and coding snippets to test user limits on previously taught topics.



Study Planner

Formulates real-time calendars and adapts daily schedules based on subject weak-points.

Intelligent Core User Journey



Robust Core Stack Integration

PLATFORM COMPONENT	INTEGRATION PURPOSE	DIRECT STUDENT BENEFIT
Mastra Orchestrator	Workflows, routing engine, and collaborative multi-agent alignment.	Seamless transition from reading context to taking quizzes or scheduling.
Qdrant Vector DB	Semantic memory storage across Student, Quiz, and Weakness collections.	High-accuracy, contextual learning that resumes exactly where they left off.
Enkrypt AI Safety	Real-time evaluation of hallucination, toxicity, bias, and safe scoring.	Guaranteed high-quality educational material safe for young minds.

Technical Objectives & Metrics

99.9%

Uptime & Educational Safety Goal

System Performance KPIs

Performance Target: Sub-5 second end-to-end response times including safety scans.

Educational Safety: Zero high-risk hallucination outputs delivered to students.

Success Indicators: Improved user retention metric, quiz accuracy gains over 4-week periods, and highly scalable Postgres + Next.js architecture.

What's Next?

Roadmap for SmartTutor AI's Expansion



Phase 2: Voice Tutor



Phase 3: PDF Analyst



Phase 4: Multilingual



Phase 5: Career Guide

| Image Sources



https://img.magnific.com/free-vector/young-scientist-learning-online_1308-172903.jpg?sem=ais_hybrid&w=740&q=80

Source: www.magnific.com